Dialectica Counter Monads

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Abstract

Dialectica constructions distinguish two coordinates: witnesses go forwards and counterexamples backwards. This variance turns a monad acting on counterexamples into a comonad on a dialectica category, and we make that observation parametric. Given a strong monad T on the base category and an Eilenberg–Moore algebra $a\colon TL \to L$ on a lineale of truth values, we define a category $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ and show that it is the co-Kleisli category of the induced comonad on the category of single counterexamples. Monoid and compatibility conditions on T and a then supply weakening, contraction and the monoidal structure required of a linear logic modality. For the free commutative monoid monad these conditions force a to be the fold of the lineale multiplication, which explains the Diller–Nahm clause and separates the multiplicative $\otimes$ from the meet used in the two-valued case. The reader and multiplicity monads, and the distributive law between them, give the three known modalities as instances, the indexed reader–multiset composite among them.

1 Introduction

The Dialectica constructions [5] model linear logic: $\mathrm{Dial}(\mathcal{C})$ the intuitionistic fragment, $\mathrm{Gir}(\mathcal{C})$ the classical one, with a decidable subcategory of $\mathrm{Gir}(\mathcal{C})$ $*$-autonomous. An object is a triple (U, X, α) — an object U of witnesses, an object X of counterexamples, and a relation α between the two, $\alpha\colon U \times X \to L$ for a lineale L — read as $\exists u \forall x\, \alpha(u, x)$; a morphism $(U, X, \alpha) \to (V, Y, \beta)$ is a pair $f\colon U \to V$ and $F\colon U \times Y \to X$ subject to the demand that $\alpha(u, F(u, y))$ implies $\beta(f(u), y)$, so that witnesses go forwards and counterexamples backwards. A model of linear logic needs a modality: the ! that restores weakening and contraction, and with them the implication of the underlying logic.

Three such modalities were described in *Dialectica Comonads* [7]. Each leaves the witness object untouched and replaces the counterexample object: $(U, X, \alpha) \mapsto (U, X^*, \widehat{\alpha})$, with X^* the free commutative monoid on X; $(U, X, \alpha) \mapsto (U, X^U, \widehat{\alpha})$, the currying comonad; and their composite, with counterexample object $(X^*)^U$. Each is a 'linear exponential comonad'[1] on a dialectica category.

Set side by side, the three share something. They are comonads on the corresponding Dialectica constructions, but in each case the data one actually supplies is a *monad* on the base $\mathcal{C}$ — the free commutative monoid, the U-reader monad, their composite — acting on the counterexample coordinate alone. Contravariance turns these monads into comonads: a morphism carries counterexamples backwards, so the unit and multiplication of a monad on X arrive as the counit and comultiplication of a comonad on (U, X, α). That this works for a generic monad is not evident in advance: a monad on $\mathcal{C}$ does not by itself extend the relation α, nor make the extension respect the morphism condition. Definition 2.2 says what more is needed, and Proposition 3.2 proves that it suffices. We abbreviate counterexample to *counter* from here on.

This note therefore varies what the backward map is allowed to return by fixing a monad T on the counter coordinate and an algebra a on the object L of truth values, and reading the relation along a. The pair (T, a) is then a choice of modality rather than data beside the other parameters of the construction: Theorem 3.3 identifies $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ with the co-Kleisli category of that modality on the category of single counters. The lineale L is fixed once and for all in §2 and its name suppressed. At $T = \mathrm{Id}$ the algebra does not disappear but is determined, the unit law forcing a to be the identity, and we write $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ for $\mathrm{Dial}_{(\mathrm{Id},\mathrm{id})}(\mathcal{C})$. The comonad laws are the algebra laws, and the linear exponential structure is a monoid on counters, compatible with the monad, together with two conditions relating a to the multiplication of L.

The instances are the three modalities of [7], and the parametrization produces no new ones. What it produces is a uniform account of them, the indexed case included, and a lineale-valued one: over the Lawvere quantale the Diller–Nahm modality charges per use of a counter, which over $\mathbf{2}$ cannot be seen.

Calculations are recalled from [7], which was written at $L = \mathbf{2}$; the pas-

[1] The terminology is Hyland and Schalk's [13]. It is not ideal — in category theory an exponential is a cartesian function space — but a name is needed for what makes ! more than a comonad, and this is the one in use. We call ! itself a modality throughout.

sage to an arbitrary lineale is described in [6] for the Girard variant $\mathrm{Gir}(\mathcal{C})$; the lineale-valued $\mathrm{Dial}(\mathcal{C})$ used below is easy. The structural calculations for $\mathrm{Dial}(\mathcal{C})$ itself are in the thesis [5].

The parametrized claim was arrived at here independently, but it has a close precedent. Powell's unpublished note of 2010 [17], which we came upon only afterwards, gives a uniform construction of comonads from strong monads for a Dialectica-style category. Bodil Biering's thesis [2] analyses Dialectica interpretations categorically. Powell's is a fibrational formulation, in which, following Hyland [14], an object is a formula in a preordered fibration. Here an object is a map $\alpha\colon U \times X \to L$ into a lineale, and the modality is supplied by an Eilenberg–Moore algebra $a\colon TL \to L$. Taking relations as maps into L is what lets L vary, and that is what makes available the parametrized category below, the forced algebra of Theorem 3.10, and the distinction between $\otimes$ and $\wedge$; asking for an algebra excludes $TX = 1$, and with it his model of modified realizability. No claim is intended that either variant is the canonical Dialectica construction.

Everything below is propositional or simply typed; type dependence is deferred. Throughout, $\mathcal{C}$ is cartesian closed and, for readability, we argue with elements, as though $\mathcal{C} = \mathbf{Set}$; Remark 2.4 says what the element-free version requires.

2 The parametrized category

Gödel's interpretation [12] assigns to a formula A a formula $\exists u \forall x\, \alpha(u, x)$ with α quantifier-free, and the categorical construction takes that shape as its objects. The relation α demands only that we know when one value entails another, that two values can be combined, and that the combination has a residual. A poset with a monoidal structure and its right adjoint supplies exactly those three, and nothing else is needed. This is what we call a lineale.

Definition 2.1. A *lineale* [16] L is a poset $(L, \sqsubseteq)$ with a symmetric monoidal structure $(\otimes, e)$ whose multiplication is monotone and has a right adjoint $\multimap$, so that $x \otimes y \sqsubseteq z$ iff $x \sqsubseteq y \multimap z$.

In the original Dialectica construction $\alpha \subseteq U \times X$, equivalently $\alpha\colon U \times X \to \mathbf{2}$. We take instead $\alpha\colon U \times X \to L$. Two things follow. The construction becomes available over bases with no subobject classifier, since all that is asked of L is that it be an object of $\mathcal{C}$ with that structure; and the multiplicative conjunction of the resulting category is computed from $\otimes$

rather than from a meet, a difference invisible at $L = \mathbf{2}$, where the two coincide. As for the backward map, the Girard variant $\mathrm{Gir}(\mathcal{C})$ takes $F\colon Y \to X$, independent of the witness, while the Dialectica construction proper takes $F\colon U \times Y \to X$; both appear below, and Remark 3.4 relates them by a comonad.

Once the backward map returns an element of TX rather than of X, the morphism condition needs a value of α at a point of $U \times TX$, and the shape of the data forces how to get one. Since α is a map into an object rather than a predicate in a fibre, we may apply T to it: the strength carries $U \times TX$ to $T(U \times X)$, and $T\alpha$ carries that to TL. Only the last step is missing, $TL \to L$, and that step explains why a must be an algebra map for the monad T, the whole of (A2). Two laws are then needed and no more. Extending α along a single counter must return α itself, or the identity morphism fails its condition; and extending twice must agree with extending once after μ, or composites fail theirs. Those are exactly the unit and associativity laws of an Eilenberg–Moore algebra, so asking for an algebra is not an extra hypothesis but the least that makes Definition 2.6 a category. Monotonicity, (A3), is what lets the resulting inequality be transported along T; it is asked of a and of nothing else.

Definition 2.2. The parameters of the Dialectica construction are a triple (L, T, a) where

(A1) T is a strong monad on $\mathcal{C}$, with unit η, multiplication μ and strength $\mathrm{st}\colon U \times TX \to T(U \times X)$;

(A2) $a\colon TL \to L$ is an Eilenberg–Moore algebra for T: $a \circ \eta_L = \mathrm{id}$ and $a \circ \mu_L = a \circ Ta$;

(A3) T is order-enriched on hom-sets into L and a is monotone: if $p \sqsubseteq q$ pointwise then $a \circ Tp \sqsubseteq a \circ Tq$.

All three of L, T and a are parameters in principle, and L especially: it is a dial of its own. Here it is fixed along with the base, and only the pair (T, a) moves; the notation records that, with the base as an argument, the pair as a subscript and L left implicit. The subscript therefore carries the modality and not the lineale, which is what it carries in $\mathrm{Dial}_2(\mathrm{Sets})$ and $\mathrm{Dial}_L(\mathrm{Sets})$; here the lineale is fixed, so the position is free, and it is used only in this note. We write $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ for $\mathrm{Dial}_{(\mathrm{Id},\mathrm{id})}(\mathcal{C})$ and call it the category of single counters; it is one member of the family and also the category the other members are built over. Three further conditions, (A4)–(A6), are needed for the modality and are stated in §3.2, where they turn out to determine

the algebra rather than merely constrain it (Theorem 3.10). The split is worth keeping: a pair (T, a) satisfying only (A1)–(A3) still gives a category, just not a model of !.

The functor T is a plain monad on the base throughout. The composite modality of $\mathrm{Gir}(\mathcal{C})$, with counter object $(X^*)^U$, needs a family of them indexed by the witness object; that variant is where the indexing of the thesis [5] belongs, and it is deferred to §5.1, where it is used.

Definition 2.3. Given $\alpha\colon U \times X \to L$, its *extension along* a is

$$\widehat{\alpha} \;:\; U \times TX \to L, \qquad \widehat{\alpha}(u,t) \;=\; a\big(T(x \mapsto \alpha(u,x))(t)\big).$$

Remark 2.4. Element-free, Definition 2.3 is where the strength is used, and the coherence of st with η and μ is what makes Lemma 2.5 go through; over a general $\mathcal{C}$ these are the conditions to spell out, and nothing else in the note needs $\mathcal{C} = \mathbf{Set}$.

What the proofs below use is the algebra laws in the form they take for the extension.

Lemma 2.5. *The extension satisfies* $\widehat{\alpha}(u, \eta x) = \alpha(u,x)$ *and* $\widehat{\alpha}(u, \mu t) = \widehat{\widehat{\alpha}}(u,t)$.

Proof. These are the unit and associativity laws of (A2), with naturality of η and μ. □

2.1 The category

Objects are untouched by the parameter (T, a): an object (U, X, α) remains the same triple. The backward map now lands in TX, and the condition compares the extension of the source relation, evaluated there, with the target relation evaluated at a plain counter. The pair acts on the source of a morphism and leaves the target alone; Theorem 3.3 is what that asymmetry is for. At $T = \mathrm{Id}$ the extension is α itself, by Lemma 2.5, and the definition below is the original one.

Definition 2.6. The category $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ has as objects triples (U, X, α) with $\alpha\colon U \times X \to L$. A morphism $(U, X, \alpha) \to (V, Y, \beta)$ is a pair

$$f\colon U \to V, \qquad F\colon U \times Y \to TX \qquad \text{such that} \qquad \widehat{\alpha}\big(u, F(u,y)\big) \;\sqsubseteq\; \beta\big(f(u), y\big).$$

Proposition 2.7. *Under (A1)–(A3),* $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ *is a category, with identity* $(\mathrm{id}, \eta \circ \pi_2)$ *and with the composite of* (f, F) *and* (g, G) *given by* $g \circ f$ *together with*

$$(u,z) \;\longmapsto\; \mu\Big(T\big(y \mapsto F(u,y)\big)\big(G(f(u), z)\big)\Big).$$

Proof. Identity: by the first part of Lemma 2.5, $\widehat{\alpha}(u, \eta x) = \alpha(u, x)$, so the condition is reflexivity. Composition: write $\bar{F}(u, s) = \mu(T(y \mapsto F(u, y))(s))$ for $s \colon TY$. By Lemma 2.5 and naturality,

$$\widehat{\alpha}\big(u, \bar{F}(u, s)\big) \;=\; a\Big(T\big(y \mapsto \widehat{\alpha}(u, F(u, y))\big)(s)\Big).$$

The hypothesis on (f, F) gives $\widehat{\alpha}(u, F(u, y)) \sqsubseteq \beta(f(u), y)$ pointwise in y, so (A3) yields $\widehat{\alpha}(u, \bar{F}(u, s)) \sqsubseteq \widehat{\beta}(f(u), s)$. Taking $s = G(f(u), z)$ and using the hypothesis on (g, G) gives the required inequality. Associativity and unitality of the composite are those of μ and η. □

The displayed identity in that proof — that the extension of α along $\bar{F}$ is the extension of the pointwise composite — is the only place where (A2) does real work, and it is where a would-be parameter that is not an algebra breaks down.

2.2 The structure of $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$

Everything §3 needs from $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is one tensor, one product, and the implication adjoint to the tensor.

Definition 2.8. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ in $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$:

$$\begin{aligned} A \otimes B &= \big(U \times V,\ X \times Y,\ \alpha(u, x) \otimes \beta(v, y)\big), & I &= (1, 1, e), \\ A \times B &= \big(U \times V,\ X + Y,\ [\alpha(u, x),\ \beta(v, y)]\big), & \top &= (1, 0, !), \\ A \multimap B &= \big(V^U \times X^{U \times Y},\ U \times Y,\ \alpha(u, F(u, y)) \multimap \beta(f(u), y)\big). \end{aligned}$$

Witnesses multiply in both conjunctions; what distinguishes them is that counters multiply in $\otimes$ and add in $\times$. The first is the multiplicative conjunction and the second the additive; $\times$ is the categorical product with $\top$ terminal, and $\otimes \dashv \multimap$. So $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is symmetric monoidal closed with finite products, a model of the multiplicative–additive fragment of intuitionistic linear logic. The thesis [5] proves this for $L = \mathbf{2}$, with objects taken as subobjects; over a lineale it is checked in [9].

Remark 2.9. The additive conjunction already gives contraction: the diagonal $U \to U \times U$ with backward map the codiagonal $X + X \to X$ satisfies its condition on the nose.

Both parameters act in the second coordinate: T replaces the counter object, a scores the result, and the witness object is carried through untouched. That is how the modality of the thesis already worked — its counter object

is the free commutative monoid on X, and the whole of the structure sits on the counters. What the parameter adds is that the monoid now comes from a monad we may choose. Monads on one side of a dialectica object, and on one side only: that is what this note is named for.

3 Counter monads as modalities

Axioms (A1)–(A3) already make $!_{(T,a)}$ a comonad: the algebra laws are the comonad laws. What they do not give is weakening, contraction and the monoidal structure, and those are (A4)–(A6).

3.1 Counter monads induce comonads

For a general pair (T, a) put

$$!_{(T,a)}(U, X, \alpha) \;=\; \big(U,\; TX,\; \widehat{\alpha}\big).$$

Proposition 3.1. *Under (A1)–(A3), $!_{(T,a)}$ is an endofunctor of $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$, sending a morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ to the pair f together with $(u, t) \mapsto T(y \mapsto F(u, y))(t)$.*

Proof. Naturality gives $\widehat{\alpha}(u, T(F(u,-))(t)) = a\big(T(y \mapsto \alpha(u, F(u, y)))(t)\big)$, and $\alpha(u, F(u, y)) \sqsubseteq \beta(f(u), y)$ pointwise in y, so (A3) bounds this by $\widehat{\beta}(f(u), t)$. Identities are preserved because $T(\mathrm{id}) = \mathrm{id}$, and composites because T is a functor. □

Proposition 3.2. *Under (A1)–(A3), $!_{(T,a)}$ is a comonad on $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$. Its counit at (U, X, α) is (id, η) and its comultiplication is (id, μ); the counit law of the comonad is the unit law of the algebra a, and the coassociativity law is the associativity law of a.*

Proof. The counit $(U, TX, \widehat{\alpha}) \to (U, X, \alpha)$ has backward map $\eta\colon U \times X \to TX$, and its condition is $\widehat{\alpha}(u, \eta x) \sqsubseteq \alpha(u, x)$, which is the first part of Lemma 2.5. The comultiplication $(U, TX, \widehat{\alpha}) \to (U, TTX, \widehat{\widehat{\alpha}})$ has backward map μ, and its condition is $\widehat{\alpha}(u, \mu t) \sqsubseteq \widehat{\widehat{\alpha}}(u, t)$, which is the second part, as an equality. Naturality in $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is naturality of η and μ with the functorial action of Proposition 3.1; the comonad equations are then those of (η, μ). □

Theorem 3.3. *The parametrized category $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ is the co-Kleisli category of $!_{(T,a)}$ on $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$.*

Proof. A co-Kleisli map $!_{(T,a)}(U, X, \alpha) \to (V, Y, \beta)$ is a pair $f\colon U \to V$, $F\colon U \times Y \to TX$ with $\widehat{\alpha}(u, F(u, y)) \sqsubseteq \beta(f(u), y)$, which is a morphism of $\mathrm{Dial}_{(T,a)}(\mathcal{C})$, and conversely. Co-Kleisli composition unfolds to the composite of Proposition 2.7. □

Two names should be kept off this. The construction here is not $\mathrm{Gir}(\mathcal{C})$, whose backward maps do not see the witness; and $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is not "the dialectica category" without qualification, since that name carries the Diller–Nahm modality with it. What $T = \mathrm{Id}$ names is the clause for implication and nothing more.

Remark 3.4 (the tower)**.** Starting instead from $\mathrm{Gir}(\mathcal{C})$, there are two comonads stacked, one for each dial. Proposition 52 of the thesis [5] gives $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C}) \simeq$ the co-Kleisli category of $\mathrm{Gir}(\mathcal{C})$ for the comonad that curries the backward map, $(U, X, \alpha) \mapsto (U, X^U, \alpha^U)$; Theorem 3.3 then gives $\mathrm{Dial}_{(T,a)}(\mathcal{C})$ as the co-Kleisli category of that for $!_{(T,a)}$. The two steps do not commute, and which order is the right one is settled in Appendix B. That the two-step picture reproduces a decomposition known independently is the best evidence available that the axioms of Definition 2.2 are the right ones.

3.2 A modality is a monoid on counters

A modality needs a monoid structure on counters, and the axioms below are read off the demand that $!A$ be a commutative comonoid; the argument is the one given informally in [7]. Ask what a comonoid in $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is. Weakening $A \to I$ has forward map $U \to 1$ and backward map $U \times 1 \to X$, that is a map $\varnothing\colon U \to X$, with condition $\alpha(u, \varnothing(u)) \sqsubseteq e$. Contraction $A \to A \otimes A$ has forward map the diagonal and backward map $U \times (X \times X) \to X$, that is a multiplication $\star\colon U \times X \times X \to X$, with condition $\alpha(u, x \star_u x') \sqsubseteq \alpha(u, x) \otimes \alpha(u, x')$. A comonoid in the dialectica category is thus a comonoid in the first coordinate and a *monoid* in the second, indexed by the first, with the relation lax between them; this is the description in the thesis [5]. The first coordinate costs nothing, $\mathcal{C}$ being cartesian, so every U is a comonoid already, and all the work is in the second: the demand is that X be replaced by something carrying a commutative monoid.

Axiom (A4) below asks for that monoid to be independent of the witness, natural in X alone. That is a restriction, and a deliberate one: a modality is meant to act the same way whatever object it is applied to, and the free commutative monoid, which is the example one reaches for first, is of this kind. What the comonoid demand adds beyond the monoid is (A5): asking

the algebra to be a monoid map is what makes the two conditions hold (Proposition 3.6).

Definition 3.5. Continuing Definition 2.2:

(A4) each TX carries a commutative monoid structure $(\varnothing_X, \star_X)$ such that

- (i) it is natural in X: every Tf preserves $\varnothing$ and $\star$;
- (ii) every $\mu_X \colon TTX \to TX$ is a map of monoids, TTX carrying the monoid at the object TX;
- (iii) $\varnothing_X$ and $\star_X$ are maps of T-algebras on the free algebra (TX, μ_X), that is

$$\mu_X \circ T(\varnothing_X) = \varnothing_X \circ !, \qquad \mu_X \circ T(\star_X) = \star_X \circ (\mu_X \times \mu_X) \circ \langle T\pi_1, T\pi_2 \rangle;$$

(A5) a is a monoid map into $(L, \otimes, e)$: $a(s \star t) = a(s) \otimes a(t)$ and $a(\varnothing) = e$;

(A6) a is lax for the pointwise multiplication: for all $w \colon T(L \times L)$,

$$a\big(T\pi_1\, w\big) \otimes a\big(T\pi_2\, w\big) \;\sqsubseteq\; a\big(T(\otimes)\, w\big),$$

and for the unit, $e \sqsubseteq a\big(T(e)(t)\big)$ for all $t \colon T1$, writing e also for the constant map $1 \to L$.

Conditions (A5) and (A6) are independent: the first relates a to the monoid $\star$ inside a single copy of TL, the second to the multiplication of L under T of a pair. The first delivers weakening and contraction, the second the lax monoidal structure.

Proposition 3.6. *Under (A1)–(A5), $!_{(T,a)}$ carries weakening and contraction:*

$$w_A \colon !A \to I, \qquad c_A \colon !A \to !A \otimes !A,$$

with backward maps $\varnothing$ and $\star$ respectively, and in both cases the required inequality holds as an equality. Each $!A$ is thereby a commutative comonoid, naturally in A.

Proof. For w_A, the forward map is $U \to 1$ and the backward map $U \times 1 \to TX$ is $\varnothing$; the condition is $\widehat{\alpha}(u, \varnothing) \sqsubseteq e$, and $\widehat{\alpha}(u, \varnothing) = a(T(\alpha(u, -))(\varnothing)) = a(\varnothing) = e$ by naturality of $\varnothing$ and (A5).

For c_A, the forward map is the diagonal and the backward map $U \times (TX \times TX) \to TX$ is $(u, (s, t)) \mapsto s \star t$; the condition is $\widehat{\alpha}(u, s \star t) \sqsubseteq \widehat{\alpha}(u, s) \otimes \widehat{\alpha}(u, t)$, and again both sides are equal, by naturality of $\star$ and (A5).

Commutativity, associativity and unitality of the comonoid are those of $(\varnothing, \star)$; naturality in A is naturality of $\varnothing$ and $\star$. □

Since $\varnothing$ and $\star$ are natural, $!_{(T,a)}$ lifts to a functor into $\mathbf{Comon}(\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C}))$; it is right adjoint to the forgetful functor. For the multiset example this recovers the familiar free-counter construction.

Three of the four ingredients of a linear exponential comonad are thus one line each. The fourth is the monoidal structure, and it is (A6) that supplies it.

Proposition 3.7. *Under (A1)–(A6), the map $m_{A,B}\colon !A \otimes !B \to !(A \otimes B)$ with forward map the identity on $U \times V$ and backward map $\langle T\pi_1, T\pi_2\rangle\colon T(X \times Y) \to TX \times TY$ satisfies its condition, and so does $m_I\colon I \to !I$, with identity forward map and the unique backward map $T1 \to 1$.*

Proof. Fix a witness pair (u, v) and a datum $w\colon T(X \times Y)$ of counter-pairs. Scoring every pair in w on both sides at once — its first component against α at u, its second against β at v — turns w into a datum of *pairs of truth values*,

$$w' \;=\; T\big\langle \alpha(u, \pi_1 -),\ \beta(v, \pi_2 -)\big\rangle(w) \;:\; T(L \times L).$$

The two sides of the required inequality are the two ways of collapsing w' to a single truth value, and naturality of T is what identifies them as such. Separating the two coordinates and folding each on its own gives $a(T\pi_1 w') \otimes a(T\pi_2 w')$, which is $\widehat{\alpha}(u, T\pi_1 w) \otimes \widehat{\beta}(v, T\pi_2 w)$, the left-hand side. Multiplying within each pair and folding once gives $a(T(\otimes)w')$, which is $\widehat{\alpha \otimes \beta}((u, v), w)$, the right-hand side. So the inequality asserts that folding separately is below folding together, which is (A6) at w'.

For m_I, the relation of $!I$ at $(*, t)$ is $a(T(e)(t))$, so the condition is $e \sqsubseteq a(T(e)(t))$ for every $t\colon T1$, the unit clause of (A6). □

Proposition 3.8. *For $T = (-)^*$ with a the fold of $\otimes$, condition (A6) holds with equality, and so it does for the composite RM of §5.1.*

Proof. For $w' = [(p_1, q_1), \ldots, (p_n, q_n)]$ the left-hand side of (A6) is $(p_1 \otimes \cdots \otimes p_n) \otimes (q_1 \otimes \cdots \otimes q_n)$ and the right-hand side is $(p_1 \otimes q_1) \otimes \cdots \otimes (p_n \otimes q_n)$; these agree by commutativity and associativity of $\otimes$. For the unit clause, $T1$ is the object of natural numbers and $a(T(e)(n))$ is the fold of n copies of e, which is e. For RM both computations hold pointwise in u. □

Coherence is not an outstanding question here. Bierman [8] showed that one equation must be added to Seely's axiomatisation [18] for it to be sound: the comultiplication must be monoidal for the structure ! carries. Here the comonoid structure is given directly by $(\varnothing, \star)$ rather than derived from $!(A \,\&\, B) \cong !A \otimes !B$, so the equation is a computation on counters.

Proposition 3.9. *Under* (A1)–(A6), $!_{(T,a)}$ *is a linear exponential comonad in the sense of Bierman [8]:* $(!_{(T,a)}, m, m_I)$ *is a symmetric monoidal comonad, each* $!A$ *is a commutative comonoid naturally in* A*, weakening and contraction are monoidal and are maps of coalgebras, and* δ *is a map of comonoids.*

Proof. The equations to be checked are between morphisms, hence between their forward and backward maps, and none of those involves L; so they hold over any lineale once the maps exist, which Propositions 3.2, 3.6 and 3.7 ensure. Forward maps are identities and projections throughout, so every condition is an equation in the counter coordinate. Write $c_{X,Y} = \langle T\pi_1, T\pi_2\rangle \colon T(X \times Y) \to TX \times TY$ for the backward map of $m_{A,B}$. Naturality of m and its associativity, unit and symmetry equations are equations between T of product projections, so they hold by functoriality of T. Monoidality of $\varepsilon = (\mathrm{id}, \eta)$ is $c_{X,Y} \circ \eta_{X\times Y} = \eta_X \times \eta_Y$, which is naturality of η at π_1 and π_2. Monoidality of $\delta = (\mathrm{id}, \mu)$ is

$$c_{X,Y} \circ \mu_{X\times Y} \;=\; (\mu_X \times \mu_Y) \circ c_{TX,TY} \circ T(c_{X,Y}) \;:\; TT(X \times Y) \to TX \times TY,$$

whose first component on the right is $\mu_X \circ T\pi_1 \circ T(c_{X,Y}) = \mu_X \circ TT\pi_1$ and on the left is $T\pi_1 \circ \mu_{X\times Y}$; these agree by naturality of μ at π_1, and the second component likewise at π_2. That weakening and contraction are monoidal asks that $\varnothing$ and $\star$ commute with c, which is their naturality at π_1 and π_2, clause (i) of (A4). That δ is a map of comonoids asks $\mu \circ \varnothing_{TX} = \varnothing_X$ and $\mu(s \star t) = \mu s \star \mu t$, that is, that μ be a map of monoids; this is clause (ii) of (A4). That weakening and contraction are maps of coalgebras asks, on backward maps, exactly the two equations of clause (iii). Proposition A.1 checks (ii) and (iii) for M and RM. The nullary conditions hold automatically: a morphism out of I has its backward map into 1, so two with the same forward map agree. □

Theorem 3.10. *Let* $T = (-)^*$ *with its free commutative monoid structure, and let* a *satisfy (A2). Then (A5) holds if and only if* a *is the fold of* $\otimes$*:* $a[x_1, \ldots, x_n] = x_1 \otimes \cdots \otimes x_n$ *for every finite multiset* $[x_1, \ldots, x_n]$*, and* $a[\,] = e$*.*

Proof. The unit law of (A2) gives $a[x] = x$ on singleton multisets. Every finite multiset is the union of its singletons and $\star$ is union, so (A5) determines a on every nonempty multiset by induction on its size, and on the empty multiset directly. The value does not depend on the order in which the singletons are taken, because $\otimes$ is commutative and associative. Conversely, the fold of $\otimes$ is a monoid map, and it satisfies (A2), since folding a union of multisets agrees with folding the folds. □

Corollary 3.11. *The algebra is not a free parameter: once T is the multiset monad, a is the lineale's own multiplication, and $\bigwedge$ is the instance $\otimes = \wedge$.*

Proof. Immediate from Theorem 3.10, the fold of $\wedge$ being $\bigwedge$. □

So $\bigwedge$ is not the canonical choice with $\otimes$ a curiosity; it is the specialisation of $\otimes$ to lineales whose multiplication is the meet, which is why it is what one sees over $\mathbf{2}$.

4 Cartesian closure

The function space of the co-Kleisli category is not a new construction. A morphism $A \to B$ there is a morphism $!A \to B$ of $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$, so the object that ought to represent those morphisms is the internal hom of Definition 2.8 evaluated at $!A$.

Definition 4.1. The co-Kleisli function space is the internal hom $!A \multimap B$ of $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$, namely $\big(V^U \times (TX)^{U\times Y},\ U \times Y,\ \gamma\big)$ with

$$\gamma\big(\langle f, F\rangle, (u, y)\big) \;=\; \widehat{\alpha}\big(u, F(u, y)\big) \multimap \beta\big(f(u), y\big).$$

Read the three coordinates in turn. A witness is a pair $\langle f, F\rangle$, which is exactly what a co-Kleisli morphism $A \to B$ consists of. A counter is a pair (u, y): a witness to try the morphism on, together with a counter for the target. And the relation is the morphism condition itself, with the lineale's $\multimap$ in place of the implication — the pair is refuted at (u, y) to the degree that it fails there.

Cartesian closure now needs one thing the axioms do not supply. Counters add in $A \times B$ and multiply in $A \otimes B$, so $!$ can turn the additive conjunction into the multiplicative one only if $T(X + Y) \cong TX \times TY$. The comparison that must be invertible runs from the product: two separate data are pushed into $X + Y$ along the injections and joined, and the joining has to be the monoid of (A4).

Theorem 4.2. *Assume* (A1)–(A6)*, and suppose that the comparison map*

$$\sigma_{X,Y} \colon TX \times TY \to T(X + Y), \qquad (s, t) \;\longmapsto\; T(\mathrm{inl})(s) \star T(\mathrm{inr})(t),$$

together with the map $T0 \to 1$, are isomorphisms; call this the splitting condition *(S). Then the co-Kleisli category is cartesian closed:*

$$\mathrm{Dial}_{(T,a)}(C \times A,\ B) \;\cong\; \mathrm{Dial}_{(T,a)}(C,\ !A \multimap B), \qquad \textit{naturally.}$$

Proof. Write $C = (W, Z, \gamma)$ and $A = (U, X, \alpha)$.

Step 1: $!(C \times A) \cong !C \otimes !A$. Both objects have witness object $W \times U$, so take the identity there and σ on counters. Write $\theta = [\gamma(w,-),\ \alpha(u,-)]\colon Z + X \to L$ for the relation of $C \times A$ at the witness (w,u); it scores a counter of the product by asking which conjunct it attacks and consulting that one.

Now $\sigma(s,t)$ is the datum holding s on the left summand and t on the right, so scoring it against θ scores s against γ and t against α, and (A5) turns the $\star$ into the $\otimes$ of the two scores. In symbols,

$$\begin{aligned}\widehat{\gamma \& \alpha}\big((w,u),\sigma(s,t)\big) &= a\Big(T\theta\big(T(\mathrm{inl})s \star T(\mathrm{inr})t\big)\Big)\\ &= a\big(T\theta\, T(\mathrm{inl})s\big) \otimes a\big(T\theta\, T(\mathrm{inr})t\big)\\ &= \widehat{\gamma}(w,s) \otimes \widehat{\alpha}(u,t).\end{aligned}$$

The first step is the definition of σ; the second is naturality of $\star$ followed by (A5); the third is $\theta \circ \mathrm{inl} = \gamma(w,-)$ and $\theta \circ \mathrm{inr} = \alpha(u,-)$. The condition holds as an equality, so the pair built from σ^{-1} is a morphism as well and the two objects are isomorphic. The nullary case is the same computation: $!\top$ has counter object $T0 \cong 1$ and relation $a(\varnothing) = e$ by (A5), so $!\top \cong I$.

Step 2: the adjunction. Four steps, of which only the middle one is Step 1:

$$\begin{aligned}\mathrm{Dial}_{(T,a)}(C \times A, B) &= \mathrm{Dial}_{\mathrm{Id}}\big(!(C \times A), B\big)\\ &\cong \mathrm{Dial}_{\mathrm{Id}}\big(!C \otimes !A, B\big)\\ &\cong \mathrm{Dial}_{\mathrm{Id}}\big(!C, !A \multimap B\big)\\ &= \mathrm{Dial}_{(T,a)}\big(C, !A \multimap B\big).\end{aligned}$$

The first and last are Theorem 3.3, the second is Step 1, and the third is $\otimes \dashv \multimap$ in $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$. Each is natural in C and in B, so the composite is. $\square$

Condition (S) is a hypothesis on T beyond (A1)–(A6). It holds for $T = (-)^*$ and for the composite RM of §5.1, and fails for the U-reader monad alone; the computations are in Appendix B. It is stated with a map rather than as an abstract splitting of T over a coproduct because it is the monoid of (A4) that supplies the comparison, and that is what makes the relations match in Step 1.

Step 1 is Girard's isomorphism $!(A \& B) \cong !A \otimes !B$ [11]. Here it is not an axiom but a computation about the monad, available because counters add where witnesses multiply.

The multiplicative structure stays on $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$. Carrying it up to the co-Kleisli category fails, and not for want of a better lineale: define the

backward map of $(f, F) \otimes (g, G)$ by the double strength, take $L = \mathbf{2}$, so that $\otimes = \wedge$ and every distributivity one might ask for holds, and let $F(u, x') = [\,]$ while $G(v, y') = [y_0]$. The product multiset is empty, so the left-hand side of the required inequality is $a[\,] = e = \mathsf{true}$ while the right-hand side is $\alpha'(f(u), x') \wedge \beta'(g(v), y')$, which may be false. What Theorem 4.2 gives is cartesian closure.

5 The monads already there

The three modalities of [7] are three settings of the parameter. The point is that a list of cases becomes a list of monads.

Example 5.1 (single counters)**.** Take $T = \mathrm{Id}$ and $a = \mathrm{id}$. Axioms (A1)–(A3) hold trivially and ! is the identity comonad; $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$ is its own co-Kleisli category. In **Set**, and in the intended nondegenerate bases, it fails (A4) at the first hurdle: $TX = X$ admits no natural $\varnothing_X \colon 1 \to X$, so there is nowhere to put the empty datum. The identity comonad is a comonad and not a modality, which is the categorical shadow of the fact that Gödel's clause for implication returns a single counterexample and buys its contraction from decidability of the atoms rather than from a !.

Example 5.2 (multisets)**.** Take $TX = X^*$, the free commutative monoid; this is the Diller–Nahm setting [10]. (A4) is that structure, with $\varnothing$ the empty multiset and $\star$ union; by Theorem 3.10 the algebra is forced to be the fold of $\otimes$, so

$$\widehat{\alpha}(u, m) \;=\; \bigotimes_{x \in m} \alpha(u, x),$$

and the morphism condition is $\bigotimes_{x \in F(u,y)} \alpha(u, x) \sqsubseteq \beta(f(u), y)$. Over a lineale with $\otimes = \wedge$, in particular over $\mathbf{2}$, this is the familiar Diller–Nahm condition. (A6) holds by Proposition 3.8 and (S) holds, so all four ingredients are present and the co-Kleisli category is cartesian closed.

Example 5.3 (the Lawvere quantale)**.** Take $L = ([0, \infty], \geq)$, with $\otimes = +$, $e = 0$ and $y \multimap z = \max(z - y, 0)$, as in Lawvere [15]. Here 0 is the most true value, a relation $\alpha(u, x)$ measures how far the witness u is from beating the counter x, and the morphism condition says that the defect of the target at y is at most the defect of the source at the counter F sends it to. The meet is max and the multiplication is $+$, so the two readings of conjunction, which coincide over $\mathbf{2}$, come apart.

For $T = (-)^*$, Theorem 3.10 forces

$$\widehat{\alpha}(u, [x_1, \ldots, x_n]) \;=\; \alpha(u, x_1) + \cdots + \alpha(u, x_n),$$

and a morphism of the co-Kleisli category bounds the defect of the target at y by the total defect of the source at the counters F proposes. A counter used n times is charged n times. Nothing else is available: with max in place of the sum, contraction would ask $\max(s \cup t) \geq \max(s) + \max(t)$, which fails as soon as two defects are positive. So over the Lawvere quantale the Diller–Nahm modality charges per use, and this is forced rather than chosen. Since $T = (-)^*$, the splitting condition holds exactly as in Appendix B, independently of the lineale, so by Theorem 4.2 the co-Kleisli category is cartesian closed.

5.1 Indexed counter monads

Everything so far has T a plain monad on $\mathcal{C}$. The composite modality of $\mathrm{Gir}(\mathcal{C})$ does not fit that: its first factor sends counters to X^U, and so depends on the witness object of the very object it is applied to. The indexing it wants is in the thesis [5], and it is confined here to the one place it is used. Replace (A1)–(A4) by their indexed forms:

(A1′) a family T_U of strong monads on $\mathcal{C}$, one for each object U, with reindexing $\varphi^* \colon T_V \Rightarrow T_U$ for $\varphi \colon U \to V$, contravariantly functorial and compatible with units, multiplications and strengths;

(A2′) a family $a_U \colon U \times T_U L \to L$, each $a_U(u, -)$ an Eilenberg–Moore algebra for T_U, with $a_U(u, \varphi^* k) = a_V(\varphi(u), k)$;

(A3′) each T_U order-enriched and each a_U monotone, as in (A3);

(A4′) each $T_U X$ carries a commutative monoid $(\varnothing_{U,X}, \star_{U,X})$, satisfying (i), (ii) and (iii) of (A4) for each T_U, and compatible with reindexing: $\varphi^*(\varnothing) = \varnothing$ and $\varphi^*(s \star t) = \varphi^* s \star \varphi^* t$.

Morphisms then have $F \colon U \times Y \to T_U X$; the composite of Proposition 2.7 reindexes $G(f(u), z)$ from $T_V Y$ to $T_U Y$ before applying $\bar{F}$, and the extra clause in (A2′) is what closes the inequality, since it turns

$$a_U\big(u,\; T_U(\beta(f(u), -))(f^* t)\big) \quad \text{into} \quad \widehat{\beta}(f(u), t).$$

The functorial action of Proposition 3.1 becomes $T_U(F(u, -)) \circ f^*$, functorial by contravariance of reindexing. The statements of §2 and §3 divide in

three. Lemma 2.5 and the equations of Proposition 3.9 are fibrewise. Theorem 3.3 retains the same description of co-Kleisli maps once the indexed comonad structure described below has been established. Proposition 3.2 and Proposition 3.6 are not fibrewise, since naturality compares the fibre over U with the fibre over V, and what makes them go through is exactly the compatibility clauses. Naturality of ε gives the backward map

$$(u, y) \longmapsto T_U(F(u,-))\big(f^*(\eta_y^{T_V})\big) \qquad \text{on one side and} \qquad \eta^{T_U}_{F(u,y)}$$

on the other, and these agree because $f^* \circ \eta^{T_V} = \eta^{T_U}$; naturality of δ compares $T_U(F(u,-))\big(f^*(\mu^{T_V} w)\big)$ with $\mu^{T_U}\big(T_U(\Phi(u,-))(f^* w)\big)$, where $\Phi(u,t) = T_U(F(u,-))(f^* t)$; expanding Φ and using naturality of μ^{T_U} leaves $f^* \circ T_V(f^*)$ against $T_U(f^*) \circ f^*$, which agree by naturality of f^*, together with the clause of (A1′) making f^* compatible with multiplications. Naturality of weakening asks $T_U(F(u,-))\big(f^*(\varnothing^{T_V})\big) = \varnothing^{T_U}$, the reindexing clause of (A4′) followed by (i); naturality of contraction asks

$$T_U(F(u,-))\big(f^*(s \star t)\big) \;=\; T_U(F(u,-))(f^* s) \star T_U(F(u,-))(f^* t),$$

the reindexing clause followed by (i) again. Those whose conditions compare two witnesses need reindexing in the statement, and there are two: Proposition 2.7, done above, and Proposition 3.7, which changes shape, since $T_{U\times V}(X \times Y) \to T_U X$ is not given by reindexing along a projection, which points the wrong way. It is given instead by reindexing along $\langle \mathrm{id}_U, v\rangle$ and $\langle u, \mathrm{id}_V\rangle$, both available because the backward map has (u, v) as an argument, so that $m_{A,B}$ has backward map

$$\big((u, v), w\big) \longmapsto \Big(T_U\pi_1\big(v^* w\big),\; T_V\pi_2\big(u^* w\big)\Big),$$

and the proof of Proposition 3.7 then runs as before, the reindexing clause of (A2′) placing both sides at $a_{U\times V}((u, v), -)$ and the required inequality being (A6) there. Condition (S), and with it Theorem 4.2, is the remaining case: it is a condition on T_U for each U separately, and Appendix B checks it for the composite RM.

Example 5.4 (currying)**.** Take $T_U X = X^U$, the U-reader monad: η is the constant map, μ is the diagonal $k \mapsto \lambda u.\, k(u)(u)$, and reindexing along $\varphi\colon U \to V$ is precomposition $X^V \to X^U$. Take $a_U(u, k) = k(u)$, evaluation. The Eilenberg–Moore laws are immediate, since $a_U(u, \eta x) = x$ and $a_U(u, \mu k) = k(u)(u) = a_U(u, T_U(a_U(u,-))(k))$, and the reindexing clause of (A2′) is $(k \circ \varphi)(u) = k(\varphi(u))$. Then $\widehat{\alpha}(u, k) = \alpha(u, k(u))$, so

$!_{(T,a)}(U, X, \alpha) = (U, X^U, \alpha^U)$ is the currying comonad. It is not a modality on its own: X^U has no natural $\varnothing$, so (A4) fails, just as for $T = \mathrm{Id}$.

What it is, is the step that moves an object out of $\mathrm{Gir}(\mathcal{C})$ and into $\mathrm{Dial}_{\mathrm{Id}}(\mathcal{C})$. Computing $!_R A \multimap B$ in $\mathrm{Gir}(\mathcal{C})$, whose homs have backward maps independent of the witness, gives witnesses $f\colon U \to V$ together with $F\colon Y \to X^U$, that is $F\colon U \times Y \to X$, and the relation $\alpha(u, F(u, y)) \multimap \beta(f(u), y)$: Gödel's clause for implication, on the nose. So the currying comonad is the modality whose co-Kleisli implication is the Dialectica one, which is the sense in which the passage from $\mathrm{Gir}(\mathcal{C})$ to $\mathrm{Dial}_{\mathrm{Id}}$ is a translation and not merely a change of morphisms.

Example 5.5 (the composite modality)**.** Take $T_U X = RMX = (X^*)^U$, the free commutative monoid applied to counters first and currying second. Now $\varnothing = \lambda u.[\,]$ and $\star$ is pointwise union, which is commutative and natural and is preserved by reindexing, so (A4) holds; $a_U(u, k) = \bigotimes k(u)$ is a monoid map, so (A5) holds; (A6) holds by Proposition 3.8; and (S) holds by Appendix B. This is the composite modality, and it is the reason this subsection exists.

Proposition 5.6. *The map $\lambda^{MR}\colon MR \Rightarrow RM$ given by $[h_1, \ldots, h_n] \mapsto \lambda u.[h_1(u), \ldots, h_n(u)]$ is a distributive law in the sense of Beck [1], so the composite RM is a monad.*

This is the law of Definition 16 of the thesis [5], the canonical map $M(X^U) \to (MX)^U$ induced by the evaluations ev_u. Its four Beck conditions and its compatibility with reindexing in U are checked in Appendix A.

Remark 5.7 (which way the law runs)**.** A distributive law in $\mathcal{C}$ and the law it induces between the corresponding comonads point in opposite directions, as recorded in [5] and again in [7], because a morphism of the dialectica category carries counters backwards. Concretely, a law $!_R!_M \Rightarrow !_M!_R$ has as its counter component a map from the counters of the target to those of the source, that is $MR \to RM$, which is λ^{MR}. So λ^{MR} is exactly what makes $!_R \circ !_M$ a comonad, and the counter object of that composite is $R(MX) = (X^*)^U$: the free commutative monoid first and currying second, as in Example 5.5. Deriving the dialectica-side law from λ^{MR}, rather than constructing it by hand, is the point of doing this in $\mathcal{C}$.

That the counter coordinate carries a monad — the identity for Gödel's interpretation, the free commutative monoid for Diller–Nahm — is in the thesis [5], as is the isomorphism $T(X + Y) \cong TX \times TY$, needed there for the one example at hand. What is added here is the step of treating the monad as a parameter, together with the lineale, so that the algebra is determined

rather than chosen (Theorem 3.10) and the indexed variant of §5.1 brings the composite modality $(X^*)^U$ inside the same account.

6 Conclusions

Making the counter monad a parameter turns three modalities into a family with more structure than the list had. The pair (T, a) is a choice of modality and the parametrized category its co-Kleisli category (Theorem 3.3), so the comonad laws are the algebra laws. For the free commutative monoid monad, (A2) and (A5) leave the algebra no freedom: it must be the fold of $\otimes$ (Theorem 3.10), of which $\bigwedge$ is the case $\otimes = \wedge$ — a statement the lineale dial makes available and that cannot be made where the truth values are simply $\mathbf{2}$. The lax condition (A6), with its unit clause, yields the monoidal structure of the modality (Proposition 3.7), and coherence then follows (Proposition 3.9). And no counter monad touches the witness object, since $!_{(T,a)}(U, X, \alpha)$ has witness object U by construction: operations on that coordinate need a different mechanism.

Everything here is propositional, or simply typed, and (A4) presumes that contraction is built from a monoid on the counter functor, which is a choice of conjunction (Remark 2.9); with the additive conjunction, counters add and the codiagonal suffices. So what is classified here are the counter monads that keep the multiplicative reading, and making the conjunction a second parameter beside (T, a) is the natural next enlargement.

Acknowledgements

Tom Powell's 2010 note [17], which I found only after the fact, overlaps this one substantially; I thank him for it. Marcelo Fiore's handwritten notes of July 2012, sent to me in 2021, contain a $\mathrm{Dial}_T(P)$ with the monad on the counter coordinate, arrived at independently; I thank him for them.

A The distributive law

Throughout, U is fixed and $R = R_U$, so that both functors are plain monads on $\mathcal{C}$ and Beck's conditions apply as stated. A distributive law of S over T is a natural transformation $\lambda\colon ST \Rightarrow TS$ with

$$\text{(D1) } \lambda \circ S\eta^T = \eta^T S, \qquad \text{(D2) } \lambda \circ \eta^S T = T\eta^S,$$

$$\text{(D3) } \lambda \circ S\mu^T = \mu^T S \circ T\lambda \circ \lambda T, \qquad \text{(D4) } \lambda \circ \mu^S T = T\mu^S \circ \lambda S \circ S\lambda.$$

The units and multiplications are $\eta^R x = \lambda u.x$ and $\mu^R k = \lambda u.\, k(u)(u)$; $\eta^M x = [x]$ and μ^M the union of a multiset of multisets.

Here $S = M$, $T = R$ and $\lambda^{MR}[h_1, \ldots, h_n] = \lambda u.[h_1(u), \ldots, h_n(u)]$.

(D1). Here $M\eta^R[x_1, \ldots, x_n]$ is a multiset of constant maps, and λ^{MR} sends it to $\lambda u.[x_1, \ldots, x_n]$, which is η^R of the original multiset.

(D2). $\eta^M h = [h]$, and $\lambda^{MR}[h] = \lambda u.[h(u)] = \eta^M \circ h = R\eta^M(h)$.

(D3). Let $w = [k_1, \ldots, k_n]$ with each $k_i \colon U \to X^U$. On the left, $M\mu^R w$ is the multiset $[\lambda u.k_1(u)(u), \ldots]$, whose image under λ^{MR} is

$$\lambda u.[k_1(u)(u), \ldots, k_n(u)(u)].$$

On the right, $\lambda^{MR}R$ gives $\lambda u.[k_1(u), \ldots, k_n(u)]$, then $R\lambda^{MR}$ gives

$$\lambda u.\lambda u'.[k_1(u)(u'), \ldots, k_n(u)(u')],$$

and $\mu^R M$ restricts to the diagonal $u' = u$. The two agree.

(D4). Let $w = [m_1, \ldots, m_k]$ be a multiset of multisets of X^U. On the left, μ^M unions them and λ^{MR} gives $\lambda u.[\, h(u) : h \in \bigcup_j m_j \,]$. On the right, $M\lambda^{MR}$ sends each m_j to $\lambda u.[\, h(u) : h \in m_j \,]$, then $\lambda^{MR}M$ gives $\lambda u.\big[[\, h(u) : h \in m_1 \,], \ldots\big]$, and $R\mu^M$ unions inside the abstraction. The two agree, union being associative.

A.1 The multiplication is a map of monoids

Proposition 3.9 asks that $\mu \colon TTX \to TX$ carry the monoid $(\varnothing_{TX}, \star_{TX})$ given by (A4) at the object TX to the monoid $(\varnothing_X, \star_X)$ on TX.

Proposition A.1. *For M and RM the multiplication μ is a map of monoids, and $\varnothing$ and $\star$ are maps of T-algebras; that is, clauses (ii) and (iii) of (A4) hold, so Proposition 3.9 applies to each.*

Proof. For M the monoid is union and μ is flattening. Flattening the empty multiset of multisets gives the empty multiset, and flattening a concatenation gives the union of the two flattenings, both on the nose.

For RM, where $TX = (X^*)^U$, the composite multiplication is

$$\mu(k)(u) \;=\; \bigcup_{g \in k(u)} g(u),$$

the diagonal in the U-reader coordinate followed by flattening, and the monoid is pointwise: $\varnothing = \lambda u.[\,]$ and $(k \star l)(u) = k(u) \cup l(u)$. Then

$$\mu(k\star l)(u) \;=\; \bigcup_{g \in k(u) \cup l(u)} g(u) \;=\; \Big(\bigcup_{g \in k(u)} g(u)\Big) \cup \Big(\bigcup_{g \in l(u)} g(u)\Big) \;=\; \big(\mu(k) \star \mu(l)\big)(u),$$

and the unit case is the empty union.

For clause (iii) with $T = M$, the map $\mu \circ T(\varnothing)$ flattens a multiset of empty multisets, which is empty; and $\mu \circ T(\star)$ unions the pairwise unions of the entries of its argument, which is the union of the first components unioned with the union of the second, that is $\star \circ (\mu \times \mu) \circ \langle T\pi_1, T\pi_2 \rangle$. For RM both clauses hold pointwise in u, from the computations above. □

A.2 Reindexing

The U-reader monad is indexed, so the law must be compatible with reindexing if the composite of §5.1 is to be available. For $\varphi\colon U' \to U$ write $\varphi^*\colon X^U \to X^{U'}$ for precomposition.

Proposition A.2. *The law commutes with reindexing:* $\lambda^{MR}_{U'} \circ M\varphi^* = \varphi^* \circ \lambda^{MR}_U$.

Proof. Here $M\varphi^*[h_1, \ldots, h_n] = [h_1 \circ \varphi, \ldots, h_n \circ \varphi]$, whose image under $\lambda^{MR}_{U'}$ is $\lambda u'.[h_1(\varphi u'), \ldots, h_n(\varphi u')]$; and $\varphi^*(\lambda u.[h_1(u), \ldots, h_n(u)])$ is the same map. □

B The splitting condition

Condition (S) asks that the comparison maps

$$T(X + Y) \to TX \times TY, \qquad T0 \to 1$$

be isomorphisms, the first being $\sigma(s, t) = T(\mathrm{inl})(s) \star T(\mathrm{inr})(t)$: the two data are pushed into $X + Y$ along the injections and joined by the monoid of (A4).

Multisets split coproducts. For $T = (-)^*$, $\star$ is union, so $\sigma(m, n)$ is the union of the images of m and n along the injections. Its inverse is restriction, $p \mapsto (p|_X, p|_Y)$: every entry of a multiset over $X + Y$ has exactly one of the two forms, so restricting a union of an inl-image and an inr-image returns the pair, and reuniting the restrictions returns the multiset. The only multiset over 0 is the empty one, so $0^* \cong 1$.

The composite inherits the splitting. For RM, with $(RM)X = (X^*)^U$,

$$\big((Z + X)^*\big)^U \;\cong\; \big(Z^* \times X^*\big)^U \;\cong\; (Z^*)^U \times (X^*)^U,$$

the first isomorphism by the multiset case applied inside the exponential, the second because $(-)^U$ preserves products; and $(0^*)^U \cong 1^U \cong 1$. This

settles the order of the two steps in Remark 3.4: the same steps the other way round give $((Z + X)^U)^*$, and the reader does not split first, so it is the free commutative monoid that must be applied to counters before the currying.

The U-reader alone does not split them. For the U-reader the condition cannot even be stated, since σ is built from $\star$ and R has none. The underlying comparison fails in any case: at $U = 1$ it would be a map between $X + Y$ and $X \times Y$, and at $X = Y = 1$ those have two elements and one.

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